

Dear Editors,

We submit "The Architectural Cost of Hybrid VTOL: meryemAircraft, a Propeller-Driven Tail-Sitting Blended-Wing-Body Without a Dedicated Lift System" for consideration as an Article in Drones, in the Drone Design and Development section.

HOW THIS WORK IS LINKED TO DRONES

The subject is an uncrewed aerial vehicle and nothing else: its configuration, its sizing, its aerodynamics, its propulsion and its control architecture. The aircraft is uncrewed by design rather than by simplification — the transition manoeuvre it performs is one a pilot cannot judge by eye, which is the recorded reason the crewed tail-sitters of the 1950s were abandoned, and the paper says so in its opening section. Its intended missions are drone missions: wildfire observation and cargo delivery to sites without a runway.

The paper's primary output is a decision tool for this journal's readership. A designer choosing between lift-plus-cruise, tilting and tail-sitting layouts for a small uncrewed aircraft has no common accounting for what each choice costs. We provide one, and we show that the ranking obtained depends on the sizing contract imposed — so a ranking quoted without its contract is not usable, and several published comparisons are quoted that way.

We note that the closest literature to this work, and six of our references, appear in Aerospace. We are submitting to Drones rather than to Aerospace because the constraints that generate every result here — uncrewed operation, small scale, no runway, an operator who is not an airline — are the constraints of this journal's subject, not of aerospace generally.

WHAT THE PAPER DOES

The paper treats the cruise-efficiency penalty of hybrid VTOL aircraft as architectural rather than as a defect of implementation, and develops it as an accounting framework in three coupled currencies: hover hardware carried through cruise, its drag when exposed, and continuous power sized by a condition holding some two percent of the flight. Every remedy we survey reduces one by raising another.

The configuration we propose addresses a specific gap. Tilting architectures satisfy the framework's escape condition — one propulsion group serving both regimes — by rotating their propulsors, and pay a pivot, its actuators, a gyroscopic moment and a transition control problem for it. Those are mechanical and control costs rather than aerodynamic ones. We propose an alternative route to the same condition: an uncrewed tail-sitting blended-wing body in which the airframe rotates and the propulsors do not. Pitch and yaw come from differential thrust between fixed-pitch propellers; roll, which coaxial torque-balanced pairs cannot produce at any setting, comes from a single variable-extension strip. What we claim to eliminate is a

mechanism class — the one that reorients a propulsor — and not every moving part; the paper is explicit about the distinction.

A second result is methodological and independent of the configuration: architectural rankings belong to the sizing contract under which a comparison is made, not to the architectures compared. We report three contracts and the ranking changes between them. We therefore claim no range advantage over the other hybrid families — we report that the tilting layout leads on range under all three contracts, and that result stands in the paper.

The case is sized at two scales and carried far enough to report what instantiating the escape condition costs, including two charges the configuration was assumed to avoid and which computation showed it does not.

WHAT WE DO NOT CLAIM

The aircraft is not shown to be flyable and we say so throughout. There is no wind-tunnel measurement and no flight test; the mass budget closes only on a battery specific power about 3.8 times the highest yet measured on a flown pack; the transition pitching moment is blocked on measurement rather than on effort; the roll strip's actuation is unsized; and the landing transition is unmodelled. The framework does not depend on that outcome, and the architectural claims above concern hardware topology rather than demonstrated performance.

AVAILABILITY AND DECLARATIONS

The computational record — mesh generator, case setups, grid-convergence study, and the scripts behind every figure — is openly available, and an archived version of this manuscript with its supplementary material is deposited at Zenodo: this version is DOI 10.5281/zenodo.22745666, and DOI 10.5281/zenodo.22144194 is the concept identifier that resolves to the latest version.

The authors have filed a patent application covering the configuration (Türkpatent application 2026/014570); this is declared in the manuscript.

The manuscript is not under consideration elsewhere and has not been published previously.

Yours sincerely,

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